

access. This is a security measure, for in its absence any app would have been able to invoke itself (essentially creating a copy itself) with elevated privileges. Hence, for an app to run as root, it must request root privileges from an external app, called the Superuser or simply 'su'. The final aim of the rooting process is to install this Superuser app on the device's internal memory as a system app. Any app that needs elevated privileges (a.k.a. root access) can then request the same from the Superuser app, a request which is usually forwarded to the user.

When an Android system boots up, the first piece of code that is executed is called the bootloader. This bootloader controls the boot process, hence access to the bootloader is required when installing a custom ROM or aftermarket firmware. Many devices, however, come with a 'locked' bootloader and need to be worked upon (read: hacked into) before we can flash a custom ROM. However, if a device has an unlocked bootloader, rooting a phone becomes relatively easy. One can either find a custom ROM for your phone model that has already been rooted and flash it onto your phone, or you can simply pull the existing ROM from your device, add the Superuser apk file to it and flash it back to your device.

Another nifty utility that Android provides is the Android Debug Bridge, better known as the ADB. The ADB allows you to execute commands entered via a computer on your Android device, essentially allowing you to VNC into your device. Any command executed on the adb shell while the phone is connected to the computer will be executed on the device. Some devices have a feature, where any command issued via the ADB is automatically given Superuser access. In this case, installing the Superuser app permanently is as simple as plugging in the phone, starting the adb and issuing the commands to copy the file to the device. The ability to execute ADB shell commands as root depends on the value of a configuration key (analogous to a registry key in Windows) called `ro.secure` whose value can be either 1 or 0. A value of 0 implies that the system will execute all commands as root user, whereas a value of 1 means apps will run with unprivileged rights. You can check the ability of your shell by entering `gettop ro.secure` into your device, either through a terminal emulator or the ADB.

Finally, in the event that both the bootloader and the ADB are locked, how does one go about rooting one's phone? This is done using vulnerabilities in low level system processes that have root access in order to interact with the device's hardware components. During the boot process, after the Linux kernel has been loaded into the memory, the init process is started